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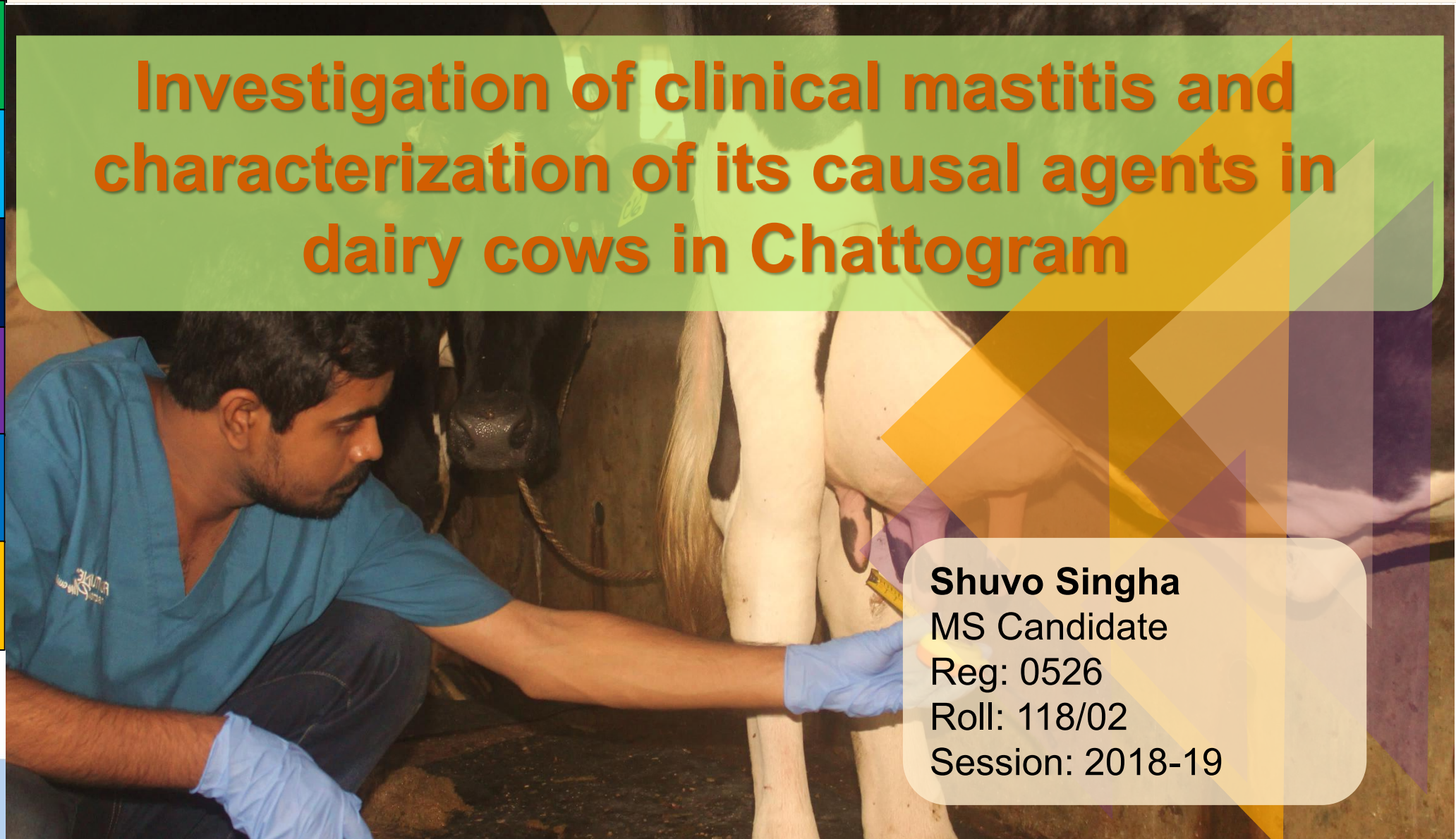
Investigation of clinical mastitis and characterization of its causal agents in dairy cows in Chattogram



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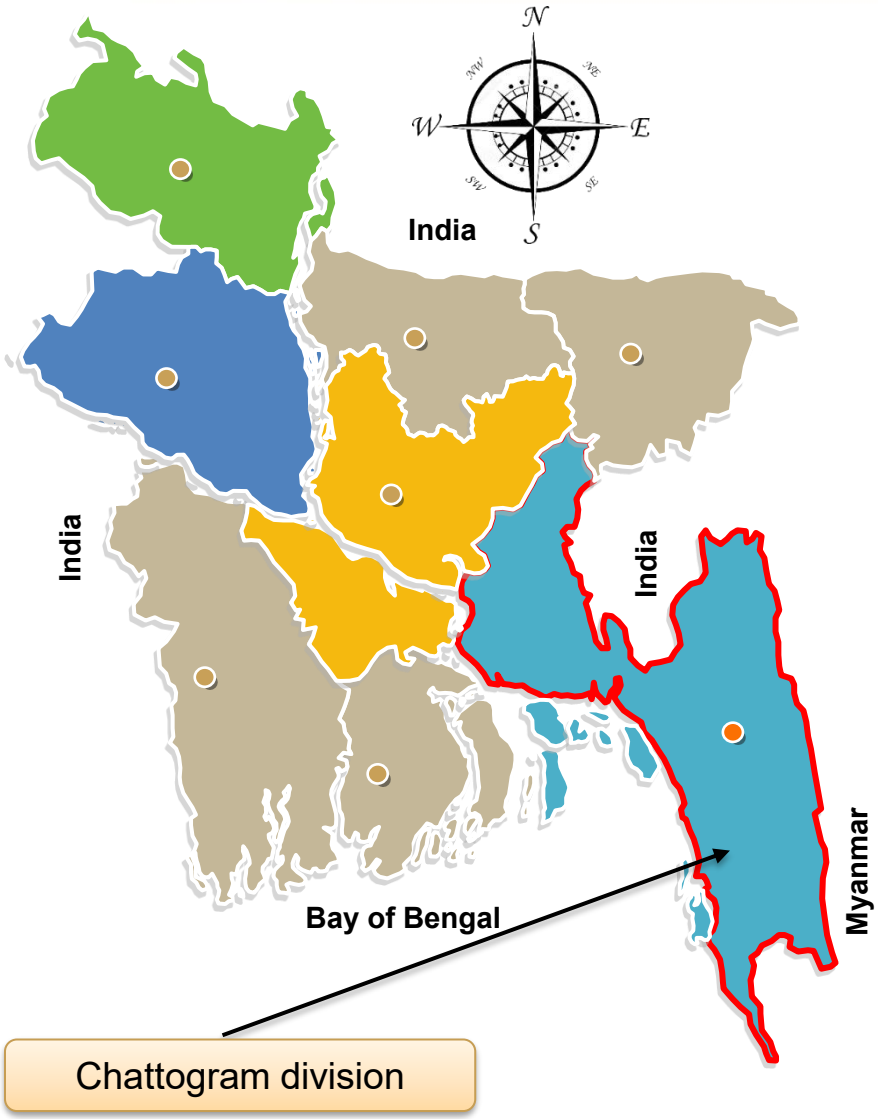
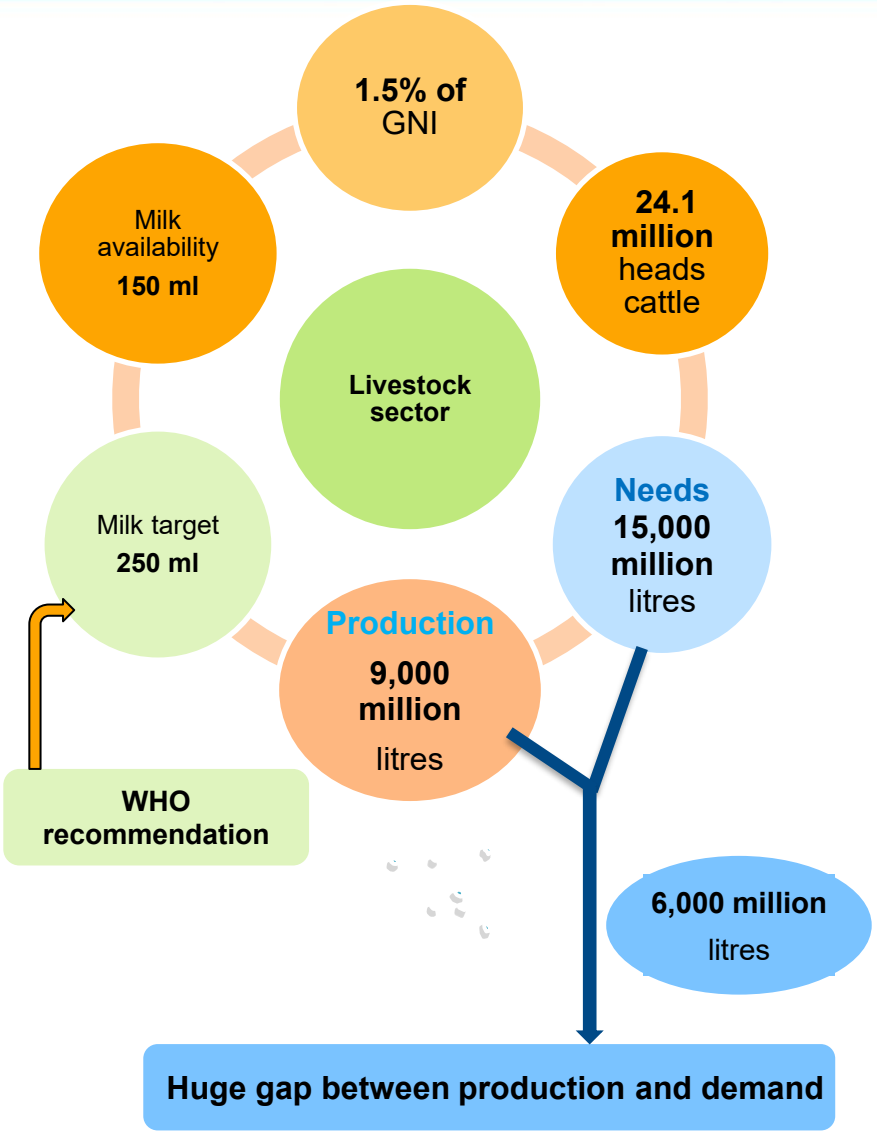
Presentation outlines

- Background
- Materials and methods
- Results and discussion
- Conclusions and recommendations
- Further studies
- Acknowledgements

Background

Population: 168.0 million
Total area: 1,47,570 sq. Km





Background

Current drivers in dairy development

To meet the gap between demand and production

Government Initiative

Recently 500 million USD investment for livestock and dairy development

Involvement into farming

Educated Youth becoming interested in dairy farming

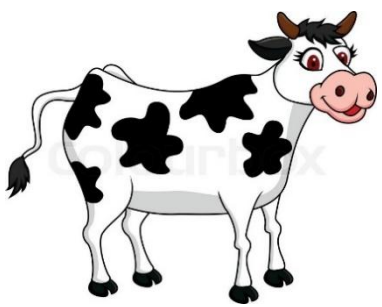
Dairy farms with better management are increasing

Background

Chattogram is one of the dairy hubs in Bangladesh

- **Inspiration from Elite people**
- **Veterinary care**
- **Govt. support**

1,551 dairy farms



2,55,744 heads

Cross breeds

- Holstein Friesian X Jersey
- Holstein Friesian X Shahiwal
- Local X Holstein Friesian
- Local X Shahiwal

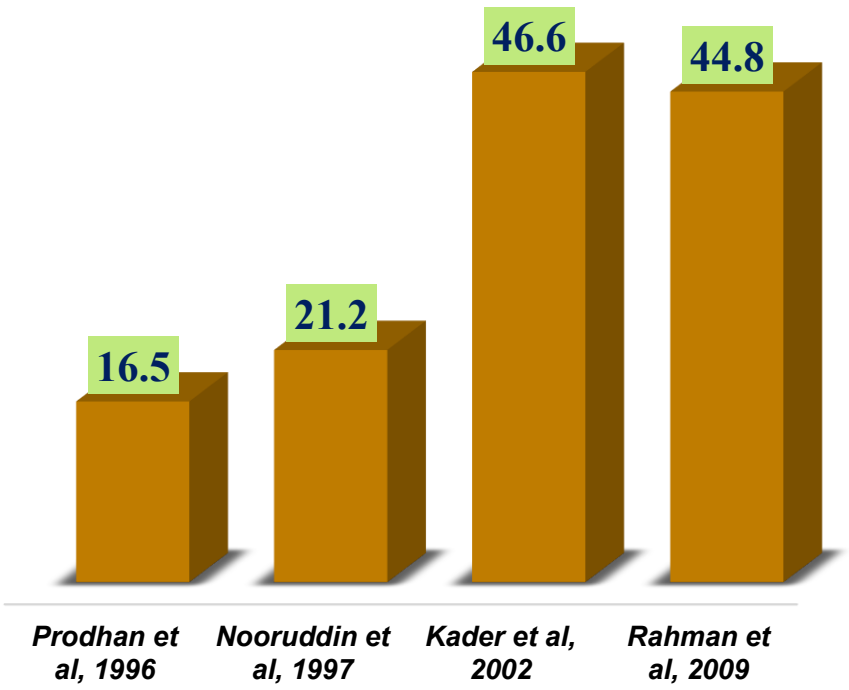
Background

Mastitis is a major challenge for dairying in Bangladesh

Challenges

- Intensive farming system
- Cross bred cows
- Fluctuation in temp. and humidity
- Feed scarcity

Mastitis picture

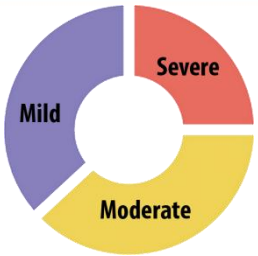


Prevalence of mastitis reports in Bangladesh: Sharma *et al.*, 2012

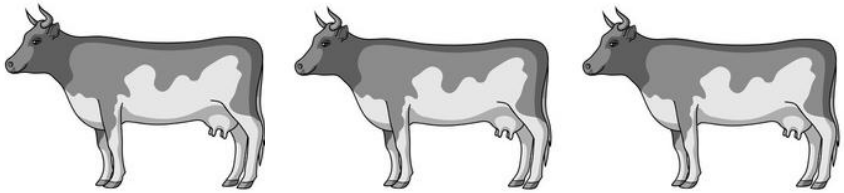
Research gaps

Few studies on clinical mastitis

Objectives



1 Clinical mastitis (CM) incidence rate and grading



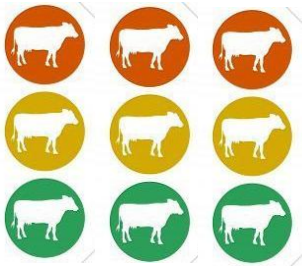
2 CM associated risk factors

3 Involved causal agents

Spin of the objectives

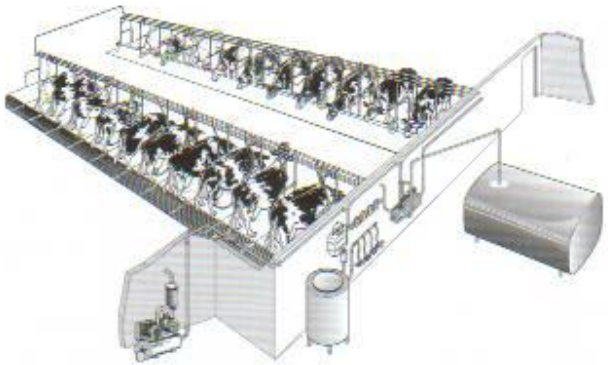
Bulk milk somatic cell count (BMSCC)

Materials and methods

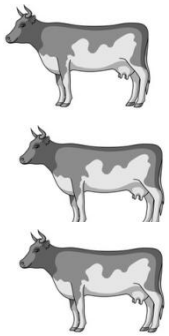


- **Lactating cows**
- Selected farms in **Chattogram** Metropoliton and surrounding areas

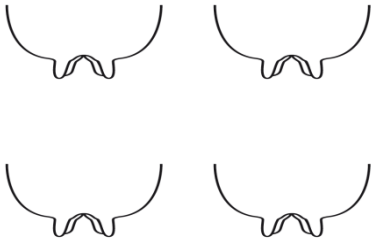
Source population



Farm



Cow



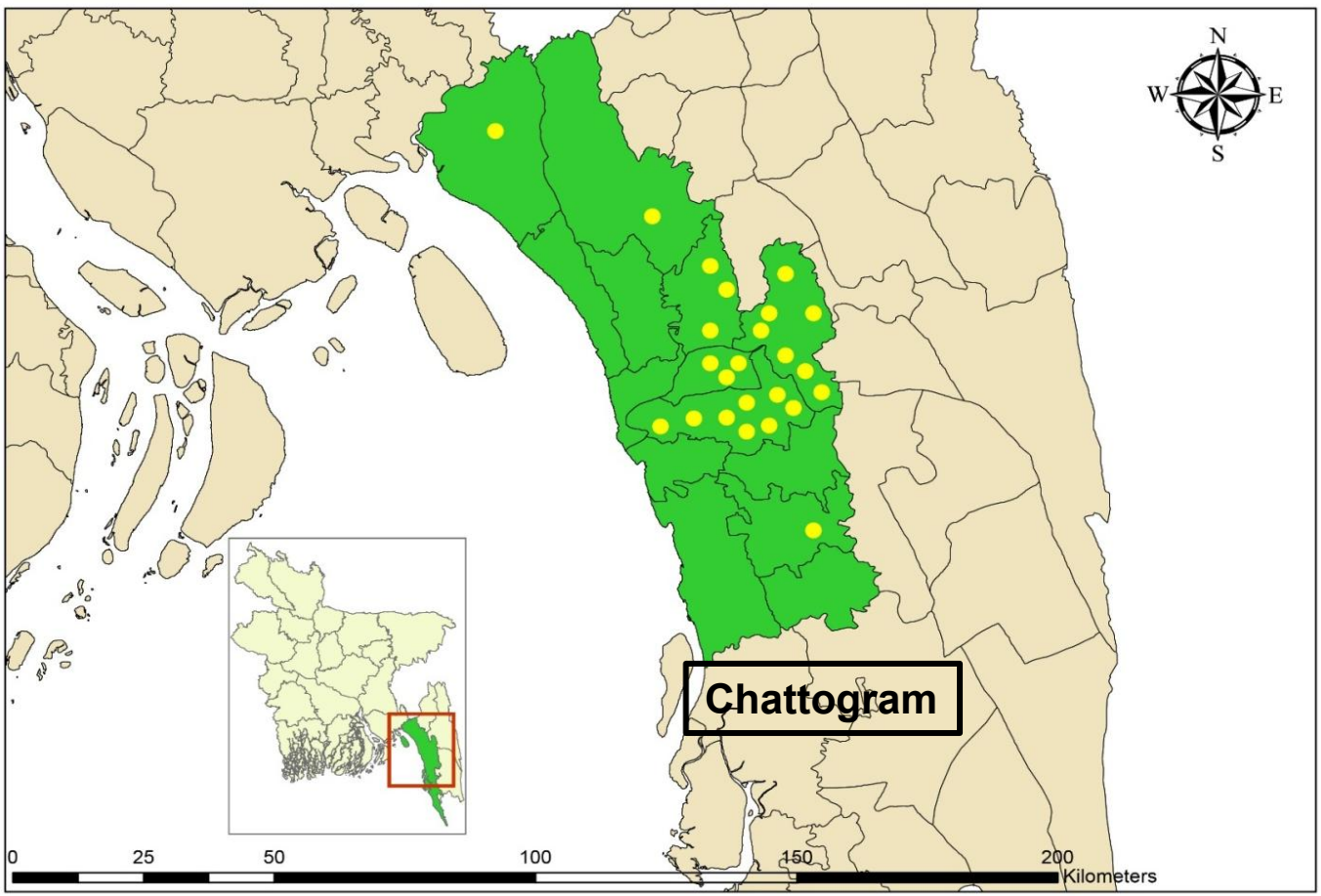
Quarter

Epi unit

Materials and methods

Sampling frame: 102 dairy farms

Location of selected 24 dairy farms under the study



Study duration

6 Months

Materials and methods

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Data Collection

Baseline sample size:
1383 lactating cows with **5,532 quarters**

Baseline data

- **Farm Data**
  - Population size
  - Pregnant/Non-pregnant animals
  - Milking animals
- **Animal Data**
  - Age, Parity, BCS
  - Current Milk Yield
  - Lactation Stage
  - Days after Parturition

Dynamic data

Entry into the study
- Pregnant heifer turned into lactating
- Newly introduced cows

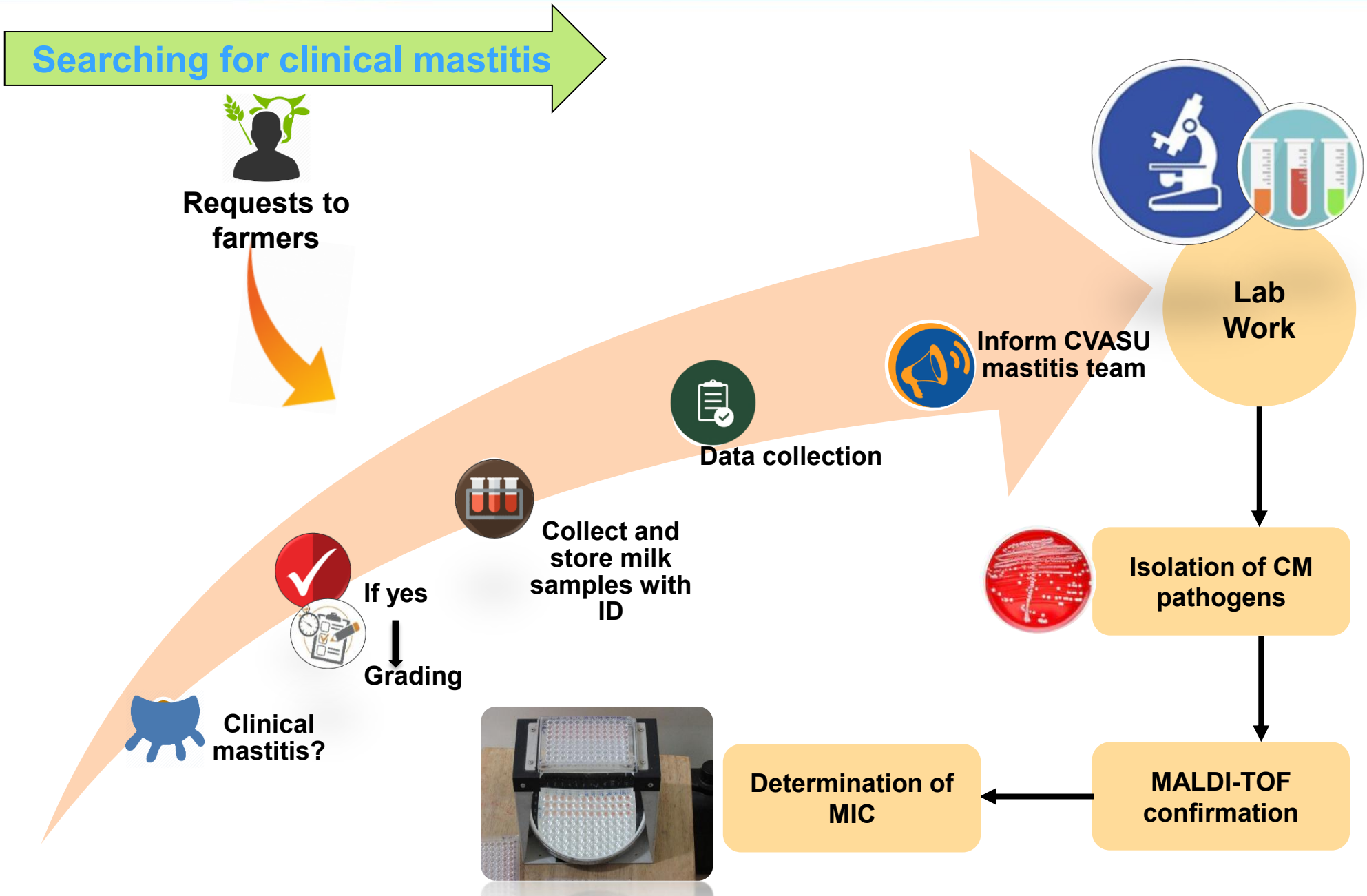
Exit from the study
- Culled lactating cows
- Dried cows
- Cases of CM

- Temperature and humidity

1st	2nd	3rd	4th	5th
June	July	August	September	October

Data were collected at one month interval

Materials and methods



Materials and methods

Mastitis Grading Record Form for Dairy Farmers

General Information

Date of recording: _____/_____/_____
Cow identification number:
Pregnant 1=Yes; 2=No
Suckled by calf 1=Yes; 2=No
Was the cow treated with antibiotics before the last dry period?
1=Yes; 2=No
If yes, how dry cow therapy is applied?

Observable characteristics of clinical mastitis

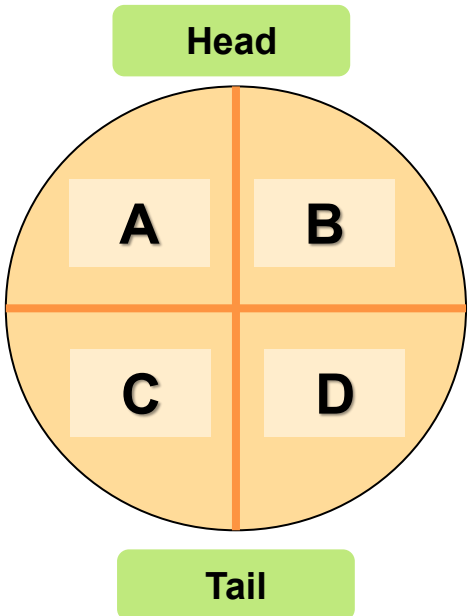
No of quarters affected: _____

Mild (Grade 1): Abnormal milk

Moderate (Grade 2): Abnormal milk accompanied by swelling or redness of mammary gland:

Severe (Grade 3): Exhibited systemic signs of illness such as depression, anorexia, dehydration, or fever:

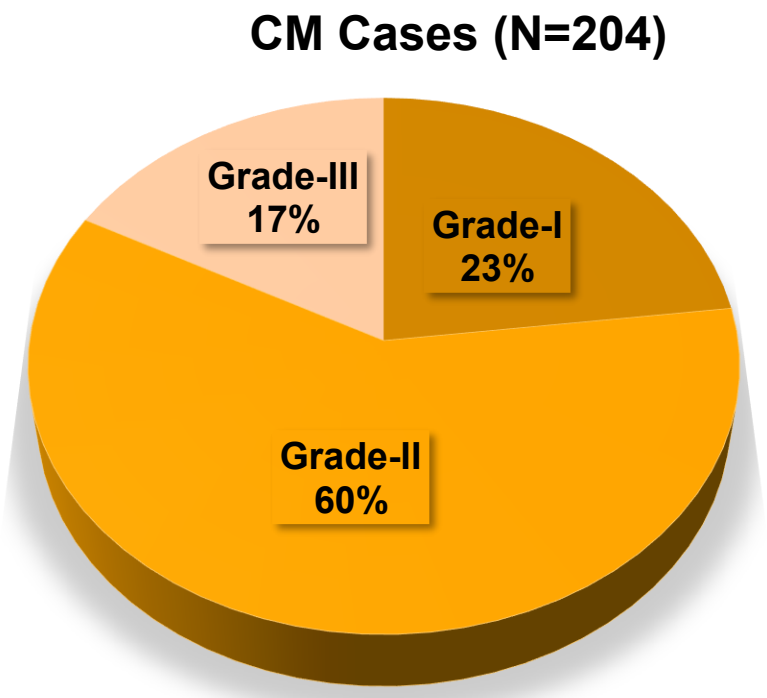
If any other udder/teat lesions, what are those?



A= Front left
B= Front right
C= Hind left
D= Hind right

Results and discussion

Clinical mastitis grading



Grade	Number of cases
I	47
II	122
III	35

Grade	Number of cases
I	63.1%
II	29.3%
III	7.0%

Verbeke et al., 2014

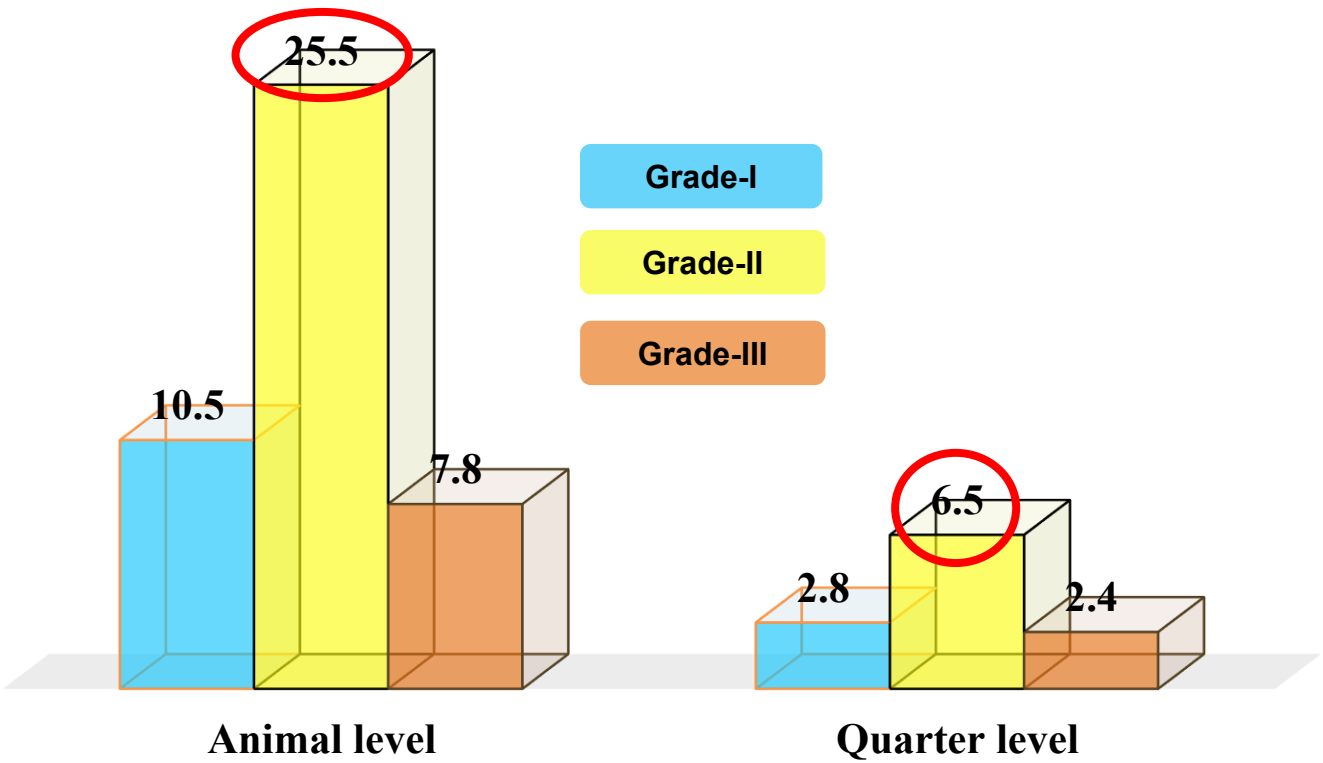
Results and discussion

Clinical mastitis incidence rate

Country	IRCM/100 cows/year
UK	47
Tanzenia	43.3
Netherlands	32.5
Canada	26.3

Bradley et al., 2007
Kivaria et al., 2007
Thompson-Crispi et al. 2003

Cow level and quarter level incidence rate at 100 unit/year



Overall

43.9 cases//100 cows/year

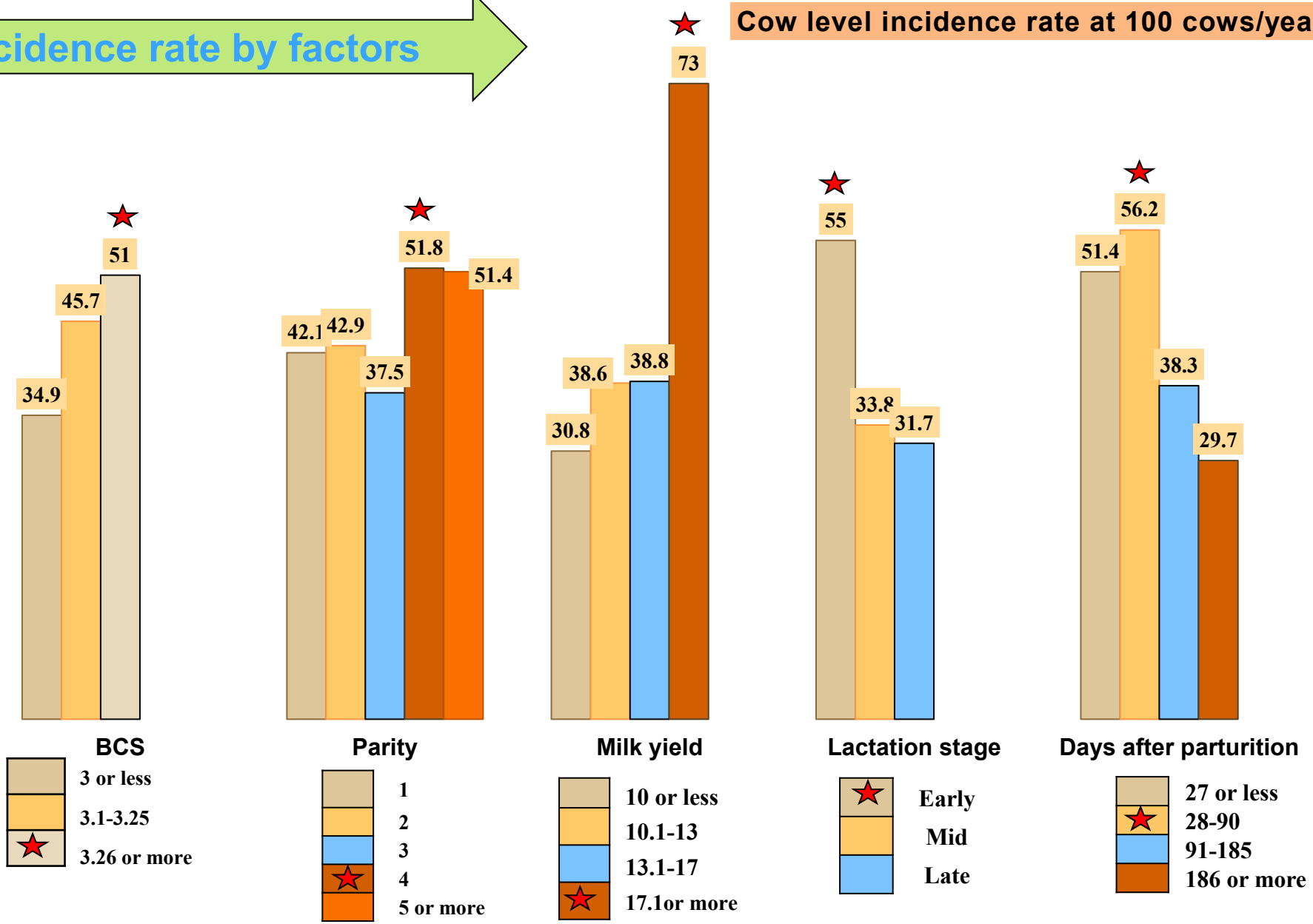
11.6 cases/100 quarters/year

Overall

Results and discussion

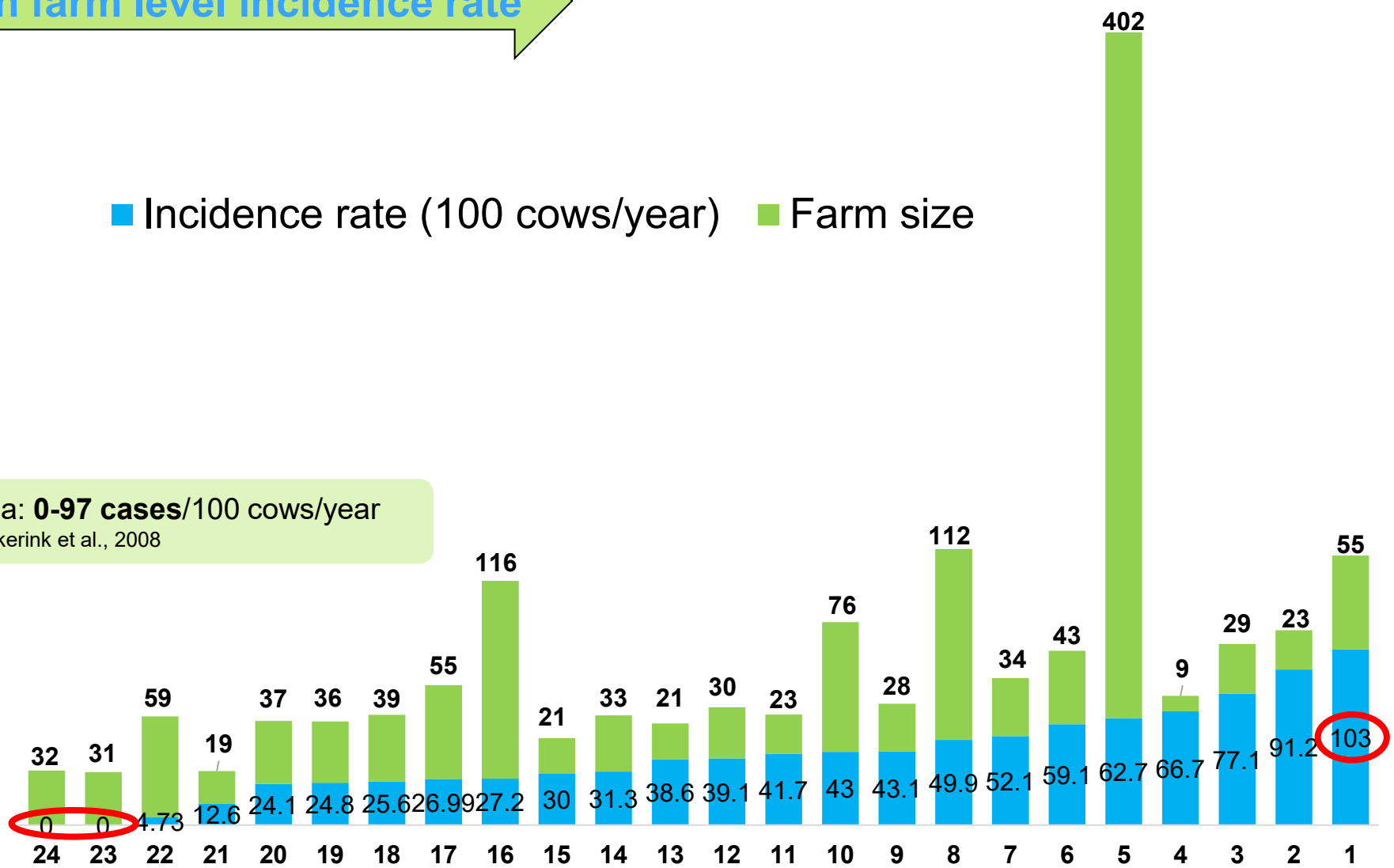
CM incidence rate by factors

Cow level incidence rate at 100 cows/year



Results and discussion

Within farm level incidence rate



Canada: **0-97 cases**/100 cows/year
Olde Riekerink et al., 2008

Results and discussion

Animal level risk factors

Multivariable logistic regression model using farm as a random effect (N=196 cases of Clinical mastitis)

Variable	Categories	N=1383	Odd Ratio	*p*-value	95% CI
BCS	3 or less	427	Ref		
	3.1-3.25	488	1.2	0.4	0.8-1.8
	3.26 or more	468	1.4	0.1	0.9-2.1
Lactation Stage	Late	352	Ref		
	Mid	283	1.1	0.7	0.7-1.8
	Early	748	1.3	0.2	0.9-2.0
Milk Yield (Liters)	10 or less	462	Ref		
	10.1-13	266	1.3	0.2	0.8-2.1
	13.1-17	328	1.4	0.1	0.9-2.24
	17.1or more	327	2.6	<0.001	1.6-4.0

Results and discussion

Bulk milk somatic cell count

Number of farms	BMSCC/ mL of milk (Lowest 50% of the farms)
24	3,54,000-7,77,500

China: 300,000
EU, Aus, NZ: 400,000
Canada: 500,000

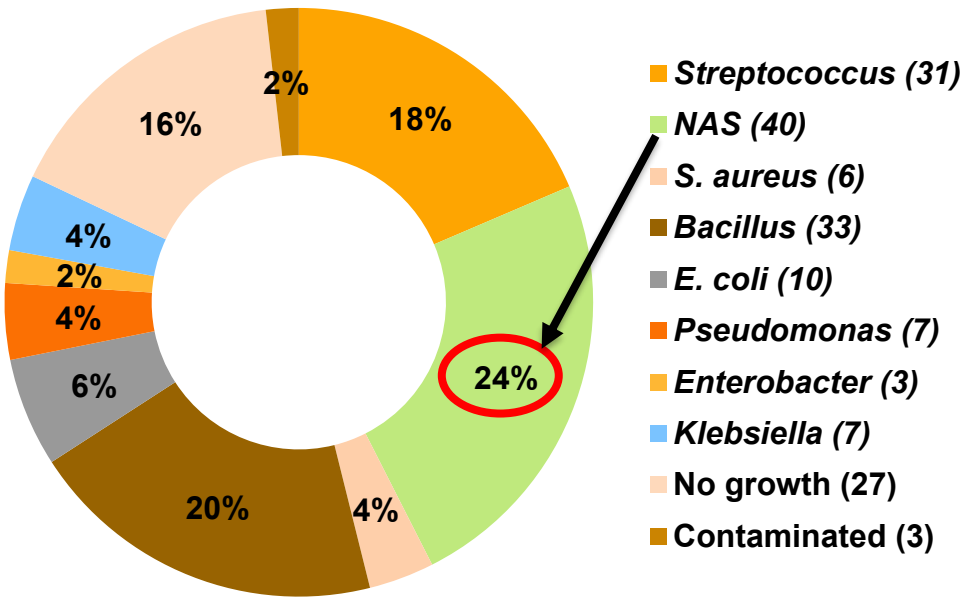
Rodrigues et al., 2005; USDA, 2013

Multivariable linear regression model using BMSCC value with the factors

Variable	Categories	Odd Ratio	*p*-value	95% CI
Source of animals	Both	Ref		
	Purchase	295.1	0.2	-130.9 to 721.1
	Own stock	-202.0	0.09	-437.4 to 33.4
Floor cleanliness	Wet	Ref		
	Dry	-244.4	0.1	-382.2 to 46.1

Results and discussion

Frequency distribution (%) of isolated bacteria (n=167)



NAS: 4.5-19.8%
Str: 18.7%
S. aureus: 0.9-12%

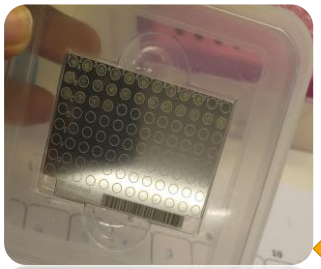
Oliveira et al., 2015;
Tomazi et al., 2018
Ndahetuye et al., 2019

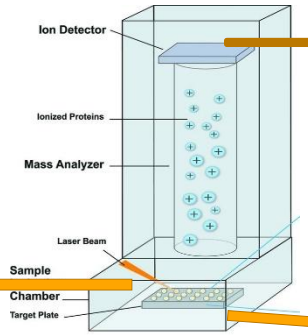
Matrix-Assisted Laser Desorption/Ionization Time of Flight Mass Spectrometry

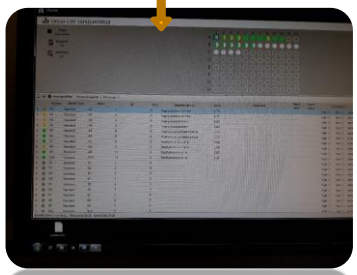
MALDI-TOF MS confirmation

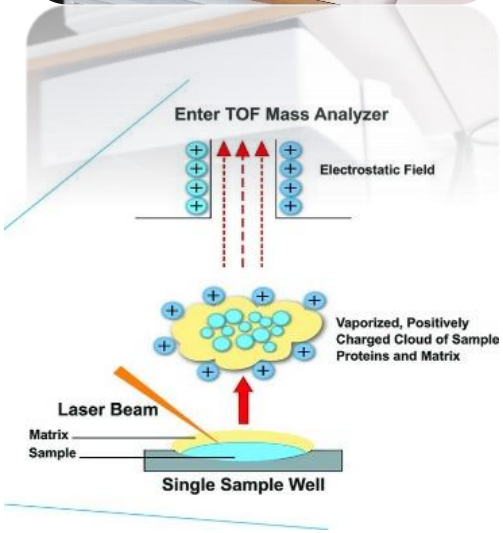
n=14 isolates

Bacteria	Number (%)
Staphylococcus aureus	2 (14.3)
Staphylococcus haemolyticus	4 (28.6)
Staphylococcus arlettae	2 (14.3)
Staphylococcus sciuri	5 (35.7)
Staphylococcus cohnii	1 (7.1)





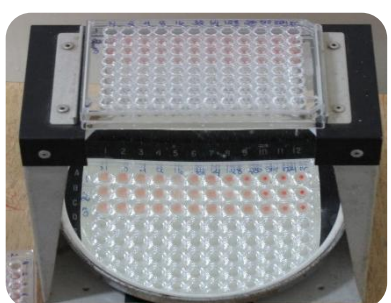




Results and discussion

Determination of minimum inhibitory concentration

Antibiotics	Isolate-1 (non-MRSA)		Isolate-2 (MRSA)	
Penicillin	S	<0.03	R	>1
Cefalotin	S	1	S	1
Oxacillin+2% Nacl	S	0.25	R	>1
Cefoxitin	S	0.25	R	>8
Enrofloxacin	S	0.25	S	0.25
Fusidic acid	S	0.5	S	0.5
Erythromycin	S	0.5	S	0.5
Clindamycin	S	0.5	S	0.5
Gentamycin	S	0.5	R	>4
Nitrofurantoin	S	16	S	16
Tetracycline	S	0.5	R	>4
Trimethoprim + Sulfonamide	S	<0.03	S	<0.03



Photos: Annica landin, SVA, Sweden

Conclusions

- High Incidence rate of CM
- BCS, lactation stage and milk yield associated with CM
- Large variation between farms
- Pathogens: NAS (24%), MRSA
- Introduction of newly purchased cows in farms correlated with higher BMSCC

Recommendations

Grading accuracy should be reached

Higher milk yielding cows with higher BCS should be managed carefully

At early lactation stage more restricted hygienic measures are necessary

Farms having higher CM incidence and higher BMSCC should adopt management following better farms

Newly purchased cows should be kept in isolation shed

Future directions

Longitudinal study covering large number of farms

Nested case control study for identifying more factors

MALDI-TOF analysis of all isolated pathogens

Determination of MIC against all isolates

Acknowledgements

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Thank you for your attention!!

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Udder Health Bangladesh

“ Healthy Udder

Safe Milk”